

ABDULLAHI MOHAMUD AHMED

Software Engineer · Full-Stack & Database Developer · Statistician & Data Analyst

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PROFESSIONAL SUMMARY

Software engineer and statistician with 7+ years designing, building and owning enterprise systems end to end. Full-stack developer across PHP, Laravel, JavaScript/jQuery and Node.js, with deep relational-database expertise in MySQL and hands-on work across PostgreSQL, Oracle and MariaDB. Currently Director of the Software Department at a Mogadishu technology company, where I own the architecture and delivery of large multi-module ERP, e-commerce, hospitality and clinical-information systems now used by 300+ businesses, clinics and hospitals across Somalia. My work spans double-entry accounting engines, hospital management systems, CRM and messaging platforms, inventory and HR modules, and a custom utility-first CSS framework. Alongside engineering I hold a BSc in Statistics and Planning and a stack of data-analytics, database and cloud certifications (Google, Meta, Oracle, Udacity), and I run an independent Somali-language NLP research programme. I combine the statistical rigour to work accurately at scale with the engineering ownership to ship and maintain production systems.

CORE COMPETENCIES

- **Full-stack engineering:** End-to-end design and delivery of enterprise web systems — architecture, backend, frontend, database and deployment.
- **Accounting & financial systems:** Double-entry posting engines, general ledger, subledgers, payroll, fixed assets, balance sheet and income-statement reporting.
- **Health information systems:** Hospital management systems covering the full patient-care lifecycle, EMR, pharmacy, laboratory, imaging, MCH and immunization tracking.
- **Database engineering:** Relational schema design, normalization, transactions, RBAC, multi-branch data scoping, and query performance across MySQL, PostgreSQL, Oracle and MariaDB.
- **Data & analytics:** Statistical analysis, data collection, validation, cleaning, deduplication, reconciliation and reporting; Python, SQL, Excel, Power BI, Tableau.
- **Integrations & messaging:** Third-party API integration — telecom SMS gateways (OAuth2), WhatsApp Business Cloud API, payment and cloud services.
- **NLP & lexicography:** Low-resource-language NLP for Somali: text-extraction pipelines, morphological processing, and structured lexicon construction.
- **Leadership & delivery:** Team leadership, code review, QA, mentoring, requirements gathering and Agile delivery.

TECHNICAL SKILLS

- **Languages & frameworks:** PHP, Laravel, JavaScript (ES6+), jQuery, Node.js, MERN stack, Python, HTML5, CSS3, Tailwind CSS, Bootstrap
- **Databases:** MySQL, PostgreSQL, Oracle Database, MariaDB — schema design, normalization, prepared statements, transactions, stored procedures
- **Data & analytics:** SQL, Python (Pandas, NumPy), Excel (advanced), SPSS, Power BI, Tableau, statistical analysis, data modeling
- **Frontend & UI:** Vanilla JS single-page apps, server-side DataTables, Chart.js, CKEditor, SweetAlert, custom utility-first CSS framework design
- **Reporting & documents:** FPDF, PhpSpreadsheet, PHPWord — PDF, Excel and Word generation; thermal/POS and A4/A5 print pipelines
- **Integrations:** RESTful API design, Hormuud Telecom SMS (OAuth2), Meta WhatsApp Business Cloud API, Guzzle, Composer, OAuth2, payment & cloud services
- **DevOps & tooling:** Git, GitHub Actions, CI/CD, Linux, Oracle Cloud Infrastructure (OCI), Windows desktop bridge apps
- **Field & data collection:** ODK (Open Data Kit) mobile data collection, data-quality assurance and cleaning workflows

PROFESSIONAL EXPERIENCE

Director of Software Department | Technology Company, Mogadishu Jan 2023 – Present

Software Developer (Mar 2019 – Dec 2022) — promoted to Director of Software Department in Jan 2023. Part-time from Mar 2019 while completing university; full-time from Jan 2020.

Own the architecture, delivery and quality of the company's enterprise product portfolio, leading engineering across e-commerce, education, hospitality, property, clinical and merchandise-ERP systems that today serve 300+ businesses, clinics and hospitals across Somalia. Progressed from sole developer to department head while remaining deeply hands-on. Selected — out of 13 candidates in a build-test exam — as the single hire from that recruitment round.

- **Leadership:** Direct and mentor a team of developers through code reviews, QA and delivery oversight; translate operational requirements from non-technical stakeholders into shipped software.
- **Ownership:** Sole technical owner of a multi-module accounting and operational-reporting platform; accountable end-to-end from architecture to deployment and ongoing maintenance.
- **Data integrity:** Lead reconciliation investigations and resolve transaction discrepancies across multiple operational data sources; design validation and deduplication processes across MySQL, PostgreSQL and Oracle.
- **Governance & security:** Apply data-governance and security practices — role-based access control, validation controls and protection of sensitive records.
- **DevOps:** Implement CI/CD workflows using GitHub Actions and OCI DevOps to improve deployment reliability and release speed.
- **Public-sector delivery:** Contributed full-stack development and relational database design to multiple public-sector systems, including World Bank-financed government platforms.

Major systems designed, built and/or owned at this company are detailed in the Key Systems & Projects section below.

Founder & Full-Stack Engineer | SomSys Technology Solutions, Mogadishu2024 – Present

Independent, registered ICT company run alongside primary employment.

- **Founder:** Founded and run a registered ICT company delivering software, web development, data analytics and network solutions.
- **Kaad PMS:** Designed and built Kaad PMS, a multi-tenant property-management SaaS product — end-to-end product design, data modeling and delivery.
- **Community projects:** Maintain the company's open-source Somali-language community and NLP projects.

Senior Consultant (ICT & Programme / Meal) | SOSCO (NGO), MogadishuJun 2025 – Apr 2026

- **Dual remit:** Engaged as a senior consultant spanning both ICT and monitoring/programme (MEAL) responsibilities.
- **ICT:** Managed organizational IT infrastructure and digital tools used for programme operations, data collection and reporting; implemented cybersecurity, data-backup and access-control practices.
- **Programme / MEAL:** Supported NGO programme reporting, data governance and operational information management across the organization.

Microsoft 365 Account Manager & ICT Consultant | Jubba Foundation (NGO), MogadishuOct 2025 – May 2026

- **M365 administration:** Administered the organization's Microsoft 365 environment — user provisioning, licensing, security policies, MFA and conditional access.
- **Training & adoption:** Delivered staff training and onboarding to strengthen digital literacy and adoption across office and field teams.
- **ICT support:** Provided additional ICT and information-management support across the foundation.

Data Collection Enumerator | DBG (local NGO), MogadishuFeb 2018 – Mar 2019

- **Field data collection:** Collected field data using the ODK (Open Data Kit) mobile app as part of a multi-person enumeration team conducting structured household interviews.
- **Data-quality assurance:** Performed daily data-quality checks — the team lead forwarded each day's collected data, which I reviewed in Excel for errors and inconsistencies and returned as validated, cleaned files.
- **Humanitarian reporting:** Supported field-level data collection, validation and reporting in a humanitarian setting.

KEY SYSTEMS & PROJECTS

The following systems were designed, built, maintained or owned during my tenure at my current employer. Per client and employer confidentiality, product and client names are withheld; systems are described by type, scope and technical content.

E-Commerce Platform & CMS (sole owner)

Full-stack web platform · PHP · MySQL · JavaScript

- Designed and built an e-commerce website and content-management system as sole owner, responsible for web design, backend, and full system reliability.
- Built accounting functionality and multiple operational reports into the CMS, enabling merchants to manage their storefronts end-to-end.

Education Management System (three product variants)

Enterprise SaaS · PHP · MySQL — SMS, IMS and UMS editions

- Maintain and enhance a multi-tenant education-management system delivered in three editions for schools, universities and small course-providing institutes.
- Resolve customer-reported issues routed via the customer-care department, fix defects and ship system improvements; still actively responsible for this product.

Construction & Property Management System

Enterprise web system · PHP · MySQL

- Maintained, improved and supported a construction and property-management system, resolving customer complaints and adding capability (product since retired by the company).

Restaurant & Hotel Management System (grew to 100+ venues)

Enterprise hospitality platform · PHP · MySQL · JavaScript · Windows desktop bridge

Took ownership of a small restaurant-management system and grew it into a large hospitality platform now used by over 100 restaurants in Mogadishu. The system has two parts — a restaurant/order-management system and a reservation office — and I added the major capabilities below.

- **Room-occupancy chart & calendar:** Built a visual room-occupancy chart showing which rooms are occupied and for how long, plus a calendar view of occupancy.
- **Automatic overstay billing:** Added automatic billing for residents who stay beyond their scheduled checkout date.
- **Inventory & menu BOM:** Built a full inventory-management system and menu bill-of-materials, tying menu items to stock consumption.
- **Deegaan Print Agent:** Designed and built a Windows print-bridge application connecting a cloud/server-hosted POS (EPOS) backend to local thermal receipt printers, so orders route automatically to the correct physical station (kitchen, bar, cashier).
- **Ongoing ownership:** Continue to maintain the platform and support its customers, with additional features shipped on a rolling basis.

Two-Part Merchandise & Clinical ERP (300+ deployments)

Large-scale ERP · PHP (mysqli) · MySQL · jQuery/AJAX · DataTables · Chart.js · FPDF · PhpSpreadsheet

I maintain the entire ERP ecosystem and personally designed and built its clinical, accounting and CRM systems. It comprises a clinic module and a merchandise module, both with full accounting. The platform is deployed at 300+ businesses, clinics and hospitals — including a major public hospital in Mogadishu. Its defining architectural idea, which I implemented, is that a patient visit/encounter is itself an invoice: every clinical order becomes a billable line item, so the clinical workflow and the revenue ledger are the same data with zero reconciliation. Detailed feature breakdown follows.

Hospital Management System (HMS) — designed & built

A complete multi-department HMS (~37,000 lines of PHP across ~120 files, plus ~26,000 lines of JavaScript), each unit with its own dashboard, patient list, tabs and reports.

- **Departments/units built:** Billing Unit; Outpatients (OPD); Pharmacy; Laboratory; Imaging; Inpatients (IPD); Mother & Child Care (MCH/Maternity); Emergency, ICU, NICU, PICU; Operations Theatre (OT); and Ward Management (room types, rooms, beds).
- **Registration & admission desk:** Single front-desk screen combining patient registration, visit creation, doctor booking and payment; autosuggest patient search, quick-registration, auto age-from-DOB, per-doctor daily visit limits, doctor rooms and specialities.
- **Automatic follow-up detection:** Checks each doctor's configurable follow-up window against the patient's last visit and applies a configured follow-up discount automatically; queue slot numbers and auto-printed admission letters (A4/A5).
- **EMR / doctor consultation:** One-page EMR: vital signs with batch saving and "last recorded" display, server-side enforcement that only the treating doctor or privileged staff can write; complaints/HPI with rich-text editing, managed diagnosis/allergy lookups, surgery history, progress and nurse notes.
- **In-consultation ordering:** Order lab tests, imaging, operations and services in-consultation, restricted by per-user JSON service catalogs; eye-care specialization (AR/IOP recording and printed reports); referrals, doctor transfer, admit-to-inpatient requests, and a full clinical-history matrix.

- **Real-time queue display board:** Standalone full-screen waiting-room TV display polling the server every second, showing the called patient (giant slot number) and waiting queue, with configurable theme colours and Somali-language headers.
- **Pharmacy & e-prescribing:** Structured Sig builder (dose/frequency/route/duration) with live stock display; Medication Administration Record (MAR) for nurse-recorded doses; checkout with per-item discounts, VAT, credit/debt toggle, amount-in-words verification and measure-aware stock decrement; unbilled-pharmacy worklist.
- **Laboratory:** Test catalog with categories, reference ranges, result styles, sub-tests and CSV bulk upload; sample-collection tracking; result-entry worklists with out-of-range flagging and document uploads; automatic SMS/WhatsApp notification to patients when results are ready; printable category-ordered reports.
- **Imaging:** Result entry with narrative findings and image/PDF attachments, unbilled-imaging worklists and printable imaging reports.
- **Inpatient care:** Admission popup usable from any unit with live room/bed availability; ward-round EMR with vitals history, medication orders + MAR, and automatic daily room/bed charges via room-type-to-product mapping; inter-department transfers; full discharge module (type/condition, DAMA, death recording, diagnoses, take-home meds, follow-up, printable summary).
- **Mother & Child Care:** Full ANC/PNC/delivery/immunization suite — antenatal visits with risk-factor checklists and printable ANC cards; delivery records with APGAR scoring and printable birth certificates; postnatal visits and immunization records; automated ANC & immunization reminders by SMS and WhatsApp in the patient's language.
- **Operations Theatre:** Operation scheduling with full clinical-team management, consent-form uploads, status workflow (Scheduled → In Progress → Completed/Cancelled), and a complete audit trail of every OT action with attachments; price-visibility privileges separated from scheduling.
- **Billing, insurance & reporting:** Receipts against any encounter, sponsor/insurance payers, credit and pay-later flows, cancellation with full transactional reversal; a reports engine (Daily, Hospital Income, Income by Department/Item/Speciality, Patient Financial Statements, Sponsor Details) filterable many ways with on-screen, CSV and PDF output; multi-format printing (thermal, A4/A5, admission letters, prescriptions).
- **Technical highlights:** ~200 granular RBAC flags per user plus unit-level access and per-user service catalogs re-checked server-side; multi-branch/warehouse scoping on every query; AJAX-first single-dispatcher architecture (~100 operations) with mysqli transactions around all multi-table writes; clinic services are inventory products, encounters are sales invoices, receipts post to the finance ledger — clinic revenue flows straight into the income statement and balance sheet.

General Journal — Double-Entry Accounting Engine — designed & built

A full general-journal module (~6,000 lines) that turned the ERP into a proper double-entry accounting system.

- **Multi-line entry editor:** Unlimited debit/credit rows with live running totals and dual client+server validation that debits must equal credits before posting.
- **Three-level chart of accounts:** Categories → detail account types → accounts, with a searchable picker; each account carries debit/credit sign operators and transaction-requirement metadata; a context-aware counterparty column that morphs per account type (client, supplier, shareholder, employee, asset, prepaid, or a UNION search for cash rows).
- **In-form master-data creation:** Create accounts, clients, suppliers, shareholders, employees, assets, prepaid expenses and bank accounts from popups without leaving the journal, including auto-provisioning of companion chart rows.
- **Config-driven posting engine:** A declarative account-type configuration maps ~18 account types to their side effects; one journal save fans out — inside a single DB transaction — to bank/cash balances and auto-generated vouchers, customer ledger with FIFO receipt allocation, vendor ledger, payroll generation, fixed-asset registration/depreciation/impairment/disposal, prepaid amortization, shareholder-equity recalculation, and an expense payment-request/approval pipeline.
- **Reverse-and-replay editing:** Editing a journal fully reverses every downstream effect (balances, vouchers, ledgers, payroll, asset schedules) via per-effect transactional undo helpers, then re-posts from the new data with full audit columns.
- **Reporting integration:** Server-side paginated, privilege- and warehouse-scoped listings feed the income statement, balance sheet and daily/bank/vendor reports, with FPDF PDF output.

CRM with SMS & WhatsApp Integration — designed & built

- **Messaging console:** KPI dashboard, SMS and WhatsApp template managers with inline status toggles and per-template “auto-send after transaction” automation, a placeholder-variable template builder with live preview, three languages (Somali Maxaatiri, Somali Maay, English) and marketing campaigns with history logging.
- **SMS gateway (Hormuud Telecom):** REST integration with OAuth2 password-grant token flow and bearer-authenticated sends; Somali phone-number normalization to +252; per-message logging; a reusable SMSManager class and a mini-ORM; business-rule guards; and bulk send from any ERP list page via a reusable modal.
- **WhatsApp (Meta Business Cloud API):** Built a self-hosted, token-protected PHP microservice (Guzzle + phpdotenv) wrapping the Meta Graph API so credentials are never embedded in the ERP; implemented template messages, OTP messages with

copy-code button, marketing templates, and document messages delivering invoice PDFs; added an OTP generator, sliding-window rate limiter and OTP storage; multilingual Meta-approved template families.

- **Event-driven automation:** Global switches plus per-template flags trigger messages on invoice creation, payment received, new client/patient, workflow/task status changes, lab-result-ready, and a cron-style scheduler for language-matched ANC & immunization reminders.

Asset Management (fixed / long-term assets) — built

- **Integrated fixed-asset module:** Asset registration with categories and useful life, bank-validated payment chains, straight-line monthly depreciation with once-per-month SQL enforcement and final-month rounding correction, loan-installment handling, guarded editing and full delete-with-reversal, plus an FPDF asset register.
- **Standalone long-term-assets application:** Richer model with asset codes, salvage value, part-payment methods, an append-only per-asset activity ledger that reconstructs the balance after every mutation, configurable depreciation policy, CapEx capitalization, impairment, and sale/disposal with automatic gain/loss — all transactional, with KPI dashboard, three reports, and general-journal and balance-sheet integration.
- **Prepaid-expense amortization:** A mirror of the depreciation pattern for prepaid expenses — registration, monthly amortization with guards, an activity ledger and journal integration.

Business Reference Setup — configurable document-numbering engine — built

- An admin feature letting each business define its own reference-number format per document type from four orderable components (counter, prefix, company/branch/country code, year/date token), stored as JSON per document type for arbitrary ordering without schema changes.
- Interactive setup UI with click-to-reorder position badges and a live preview (e.g. INV/010/SOM/2026); zero-padding-aware counter; a two-phase commit so failed saves never burn numbers; per-warehouse persistence and privilege gating; references generated on invoices, receipts and payments and printed on all FPDF templates.

HRM — Human Resource Management (two implementations) — built

- **Integrated HR module:** Employee registry with job/salary details, per-service piece rates, clinic extensions and signature uploads; versioned employment history; departments & specialties with referential-integrity guards; monthly payroll generation ($\text{Net} = \text{salary} + \text{allowance} + \text{bonus} - \text{advance} - \text{deduction}$) with full accounting integration; advance payments with configurable limits, piece-rate pay, bonuses/deductions; two-way general-journal integration; and a user/privilege administration system with bcrypt hashing and a ~80-flag rights matrix enforced server-side.
- **Standalone HRM application:** A full HR suite with its own login — complete employee files with typed document management and photo handling; company-level compensation and installment-deduction structures; configurable tax brackets; work shifts; CSV attendance import matching actual vs planned times with leave auto-flagging and finalization; leave management with approval workflow and balances; payroll from compensation rules with an automatic tax line; FPDF payslips and dossiers; and its own 26-flag privilege system.

Balance Sheet Report — built end-to-end

- Computed live from ~25 operational subledger tables (no traditional trial balance — derived directly from operational data), with two modes: the open financial year computed live and closed years read from snapshot tables written by a year-close routine I built.
- Full assets and liabilities/equity derivation (receivables/payables netted from activity ledgers, inventory and WIP valuation, long-term assets net of accumulated depreciation, VAT/dividend/salary payables, per-shareholder equity), a derived net-income engine, a self-balancing “Open Balance” plug, multi-currency normalization, and three output channels (AJAX HTML with drill-down, FPDF PDF, PhpSpreadsheet Excel).

UI Redesign — the “ed-styles” design system — built

- **Custom utility-first CSS framework:** Authored ed-styles.css, a programmatically generated Tailwind-style utility framework (~60,000 rules, ed- prefixed) with 1-pixel-step spacing/sizing, a full color system (~44,000 rules with generated shade ramps around a dark-teal #015958 primary), button variants and styled controls.
- **Redesigned application shell & dashboard:** A new collapsible icon sidebar with inline SVG icons, flyout submenus and localStorage-persisted collapse state; a new header with notification dropdown and profile menu; and a redesigned dashboard with Chart.js v4 widgets (net-income bar chart, expenses doughnut with center-total) fed by AJAX.
- **Incremental migration:** Rolled the framework out across 188 PHP template files (~16,900 class usages) spanning every domain, shipping alongside the legacy 13,000-line stylesheet so the redesign migrated screen by screen without breaking the live product.

Cross-cutting engineering themes

- Unified data model where clinical encounters, sales invoices and the finance ledger are one system, eliminating operations-to-accounting reconciliation.

- Config-driven double-entry posting engine fanning out to 10+ subledgers inside single DB transactions, with full reverse-and-replay edit semantics.
- ~200-flag role-based access control with per-user sections, per-service catalogs and multi-branch/warehouse scoping enforced on every query.
- Third-party API integration: Hormuud Telecom SMS (OAuth2) and Meta WhatsApp Business Cloud API via a self-built, token-protected PHP microservice with rate limiting and OTP flows.
- End-to-end multilingual messaging (Somali Maxaatiri, Somali Maay, English) driven by per-client language.
- Financial reporting across three output channels (AJAX HTML, FPDF PDF, PhpSpreadsheet Excel).

INDEPENDENT PROJECTS & RESEARCH

Qaamuuska — Somali Digital Dictionary & Computational Lexicon

Full-stack NLP & lexicography · Python · PHP · JavaScript · JSON · academic research · May 2026 – Present

An end-to-end Somali-language dictionary platform and NLP research project — one of the more structured digital-lexicography efforts for Somali, a historically under-resourced language in computational linguistics.

- **Computational lexicon:** Building a structured Somali computational lexicon of 46,000+ curated headwords/entries covering definitions, part-of-speech tagging and morphological information; designed the data schema, normalization rules and quality-control workflow for large-scale lexical data.
- **Text-extraction & NLP pipeline:** Designed a Python extraction and preprocessing pipeline (`extract_content.py`) automating cleaning, tokenization and normalization of Somali orthography, plus a dedicated NLP module with morphological analysis tailored to Somali's agglutinative structure; iterated the dataset through three major versions to a production JSON store.
- **Full-stack web application:** Built a public dictionary web app — a PHP REST API serving full-text/prefix/filtered search and a vanilla-JS single-page frontend with debounced real-time search and responsive design — handling data cleaning, deduplication, database design, search and presentation, with Python migration tooling.
- **Academic paper:** Writing a formal research paper documenting the methodology, lexicon-construction approach and contribution to low-resource-language NLP, intended for a peer-reviewed computational-linguistics venue.

KGS Identity Management System

Full-stack web application · PHP · MySQL · HTML/CSS/JavaScript · LAMP

- **Identity & access platform:** Built a custom identity-management web application handling identity verification, user management and access control, from database-schema design through to a dynamic web interface.
- **Iterative schema design:** Designed and iterated the normalized relational schema through 10+ versions — users, roles, permissions and identity records with foreign-key integrity, RBAC separating authentication from authorization, audit-trail support and indexed query performance.
- **Backend & security:** PHP authentication with secure session management and bcrypt password hashing, registration/onboarding workflows, role management, identity-record CRUD, a verification state machine (pending → verified → flagged/rejected), prepared-statement SQL injection prevention and role-based route protection.
- **Frontend:** Login/registration with client-side validation, an admin dashboard with user/verification queues, profile pages with status indicators, role-assignment controls and search/filter, on a responsive CSS layout. Developed alongside a companion notary platform in the same environment.

Kaad PMS — Property-Management SaaS

Multi-tenant SaaS · via SomSys Technology Solutions

- Designed and built a multi-tenant property-management SaaS product end-to-end — product design, data modeling and delivery — under my own registered company, SomSys Technology Solutions.

EDUCATION

BSc in Statistics and Planning | SIMAD University, Mogadishu Sep 2015 – Dec 2019

- Faculty of Economics, Department of Statistics and Planning; graduated December 2019.
- Completed the university's concurrent English-language program (finished Nov 2019).

Self-Directed Software Engineering Study | Independent Jun 2018 – Present

- Structured self-study (Udemy, YouTube and hands-on projects): HTML → CSS → Bootstrap → PHP → jQuery, progressing to React, Node.js and the full MERN stack, and later Laravel and Tailwind CSS.

CERTIFICATIONS

- Google Advanced Data Analytics Professional Certificate — Coursera, 2026
- Meta Database Engineering Professional Certificate — Coursera, 2026
- Oracle Cloud Infrastructure (OCI) 2025 Developer Professional — Oracle, 2025
- Oracle Cloud Infrastructure (OCI) 2025 DevOps Professional — Oracle, 2025
- Data Analyst Nanodegree — Udacity, 2021
- WorldQuant Data Science Lab — WorldQuant University, 2026
- Delivering Quality Work with Agility (IBM Agile) — IBM, 2022
- Introduction to Cloud Computing — IBM / edX, 2022
- Introduction to Cloud Development with HTML, CSS and JavaScript — IBM / edX, 2022

LANGUAGES

- **Somali:** Native speaker; excellent written expression and orthography; deep knowledge of Somali morphology, dialect variation and vocabulary developed through original lexicographic research.
- **English:** Fluent (professional working proficiency); daily working language for documentation, client communication and research.
- **Arabic:** Intermediate.